vas.livanos@gmail.com
Website
Google Scholar

Research Interests

- Algorithms under Uncertainty
- Combinatorial Optimization
- Matroid Theory
- Algorithmic Game Theory
- Online Mechanism Design

Languages

- Greek (Native)
- English (Fluent)
- Spanish (Intermediate)
- Italian (Basic)
- French (Basic)

Vasilis Livanos

Experience

Mar 2026 - current, University of Southern California
Post-Doctoral Researcher. Host: Shaddin Dughmi

Oct 2024 - Apr 2025, EPFL
Visiting Post-Doctoral Researcher. Host: Andrés Cristi.

Sep 2024 - current, Archimedes AI
Affiliated Researcher.

Jan 2024 - Jan 2026, University of Chile & CMM
Post-Doctoral Researcher. Hosts: José Correa and José Soto.

Education

Aug 2017 - Dec 2023, University of Illinois at Urbana-Champaign
Ph.D. in Computer Science. Advisors: Ruta Mehta and Chandra Chekuri.
Thesis: Optimal Stopping Problems And Combinatorial Optimization Under Uncertainty

Oct 2011 - Mar 2017, National Technical University of Athens
Diploma in Electrical and Computer Engineering. Advisor: Dimitris Fotakis.
GPA: 9.2/10 (Top 2%) - Excellent.

Conference Publications

[15] ***Separating Measures by Hyperplanes***
with José Correa, Svenja Griesbach, El-Farouk Harb, Felix Hommelsheim, Reda Jlibene and Eilon Solan, *in submission*

[14] ***Optimal Contention Resolution from Samples***
with Shaddin Dughmi and Pranav Nuti, *in submission*

[13] ***The Strong Secretary Conjecture is True for Linear Matroids***
with Kristóf Bérczi, Shaddin Dughmi, José Soto and Victor Verdugo, *in submission*

[12] ***The Power of Sampling for Finding Far-Apart Bases in Matroids***
with with Mayank Goswami, Pasin Manurangsi, Arturo Merino, GiBeom Park and Adarsh Srinivasan, *SODA 2027*

[11] ***Threshold Dynamics and Correlated Prophet Inequalities***
with José Correa, Maximilian Fichtl, Reda Jlibene, Rida Laraki, Kevin Schewior and Victor Verdugo, *EC 2026*

[10] ***Free-Order Secretary for Two-Sided Independence Systems***
with Kristóf Bérczi, José Soto and Victor Verdugo, *in submission*

[9], ***On the Informativeness of Moments in Optimal Stopping***
with José Correa, Andrés Cristi, Victor Verdugo and Jiechen Zhang, *STOC 2026*

[8] ***Minimization I.I.D. Prophet Inequality via Extreme Value Theory: A Unified Approach***
with Ruta Mehta, *EC 2025*

[7] ***Matroid Secretary via Labeling Schemes***
with Kristóf Bérczi, José Soto and Victor Verdugo, *IPCO 2025*

[6] ***Improved Mechanisms and Prophet Inequalities for Graphical Dependencies***
with Kalen Patton and Sahil Singla, *EC 2024*

[5] ***Oracle-Augmented Prophet Inequalities***
with Sariel Har-Peled and El-Farouk Harb, *ICALP 2024*

[4] ***Minimization is Harder in the Prophet World***
with Ruta Mehta, *SODA 2024*

[3] ***Simple and Optimal Greedy Online Contention Resolution Schemes***
NeurIPS 2022

[2] ***(Almost) Envy-Free, Proportional and Efficient Allocations of an Indivisible Mixed Manna***
with Ruta Mehta and Aniket Murhekar, *AAMAS 2022*

[1] ***On Submodular Prophet Inequalities and Correlation Gap***
with Chandra Chekuri, *SAGT 2021*

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Journal Publications

[4] ***About a Ball Removal Process on Bins***
with José Correa, Marcos Kiwi, Eilon Solan and Ron Solan, *Stochastic Systems*

[3] ***Matroid Secretary via Labeling Schemes***
with Kristóf Bérczi, José Soto and Victor Verdugo, *to appear in Mathematical Programming*

[2] ***Improved Mechanisms and Prophet Inequalities for Graphical Dependencies***
with Kalen Patton and Sahil Singla, *to appear in Mathematics of Operations Research*

[1] ***On Submodular Prophet Inequalities and Correlation Gap***
with Chandra Chekuri, *Theoretical Computer Science*

Theses

2023, ***Optimal Stopping Problems And Combinatorial Optimization Under Uncertainty*****, PhD Thesis**

2017, ***Convergence and Equilibria in Coevolutionary Opinion Formation Games*****, Diploma Thesis**

Scientific Visits

Jan. 2025, 1-month visit at University of Chile
Host: Jose Correa

Sep. 2023 - Nov. 2023, 2-month visit at Georgia Institute of Technology
Host: Sahil Singla

Feb. 2023, 1-month visit at Eötvös Loránd University
Host: Kristóf Bérczi

Jan. 2023, 1-month visit at University of Chile
Host: Jose Correa

Selected Talks and Presentations

For a full list click here.

2026, Talk - ***Threshold Dynamics and Correlated Prophet Inequalities***
27th ACM Conference on Economics and Computation (EC), Rome, Italy.
Criteo AI, Paris, France.
Workshop on Optimization and Algorithms (WOA), Viña del Mar, Chile.

2025, Talk - ***Minimization I.I.D. Prophet Inequality via Extreme Value Theory: A Unified Approach***
26th ACM Conference on Economics and Computation (EC), Stanford, CA, USA.
Drexel University, Philadelphia, PA, USA.
UC Irvine, Irvine, CA, USA.

2025, Talk - ***Matroid Secretary via Labeling Schemes***
26th Conference on Integer Programming and Combinatorial Optimization (IPCO), Baltimore, MD, USA.
Summer Workshop on Combinatorial Optimization, Santiago, Chile.
SUPSI, Lugano, Switzerland.

2024, Talk - ***Improved Mechanisms and Prophet Inequalities for Graphical Dependencies***
INFORMS Annual Meeting, Seattle, USA.

2024, Tutorial - ***An Overview of Optimal Stopping Theory***
UC Irvine, Irvine, USA.

2024, Talk - ***Minimization is Harder in the Prophet World***
ACM-SIAM Symposium on Discrete Algorithms (SODA) 2024, Alexandria, USA.
EPFL, Lausanne, Switzerland.
Google, Zurich, Switzerland.
ETH, Zurich, Switzerland.
University of Bonn, Bonn, Germany.

2022, Talk - ***How to Choose a Partner Using Math***
2022 PhD Intern Conference (PIRC), Google, USA.

2021, Talk - ***On Submodular Prophet Inequalities and Correlation Gap***
14th Symposium on Algorithmic Game Theory (SAGT), Aarhus, Denmark.

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Awards and Honours

2023, *Dissertation Travel Award*, University of Illinois at Urbana-Champaign
Awarded for the completion of the PhD dissertation.

2022, *NeurIPS Scholar Award*, NeurIPS '22
Awarded to provide financial assistance to graduating PhD students.

2020, *Outstanding Teaching Assistant Award, Lifetime*, University of Illinois at Urbana-Champaign
Awarded based on ranking by students and recommendation by faculty.

2018 - 2022, *List of Teachers Ranked as Excellent*, University of Illinois at Urbana-Champaign
Ranked "Outstanding" based on students' ratings on five occasions.

2017 - 2018, *Andrew and Shana Laursen Fellowship*, University of Illinois at Urbana-Champaign
Awarded to provide meaningful assistance in the recruitment of top graduate students to the Department of Computer Science.

2012 - 2016, *John S. Latsis Foundation Undergraduate Scholarship*, National Technical University of Athens
Awarded continuously for achieving excellence during undergraduate studies.

2011, *Great Moment for Education Scholarship*, Eurobank EFG
Awarded for being first among all high school students in the national final examinations.

Professional Activities and Service

Program Committee Member
WINE '26, EC '26, WINE '25, FSTTCS '25, APPROX '25, EC '25, EC '24

Peer Reviewer

- *Journals:* SIAM Journal of Computing, Journal of Computer and System Sciences, Journal of Applied Probability, Theoretical Computer Science, Mathematics of Operations Research, Algorithmica, Mathematical Programming, Transactions in Algorithms, Operations Research Letters.
- *Conferences:* STACS '27, '24, - ITCS '27, '25, '24 - SODA '27, '26, '25, '24, '23 - STOC '26, '24, '23, '18 - IPCO '26, '24, '23, '22 - LATIN '26 - FOCS '24 - EC '22, '20, '18 - ESA '23 - SOSA '23 - ICALP '22, '21, '19 - WINE '21, '18 - WWW '18 - SAGT '18.

EC'22 Mentoring Workshop, *Mentor*

Teaching

Jan 2018 - Dec 2022, *Teaching Assistant*, University of Illinois at Urbana-Champaign

- "CS 583 Approximation Algorithms" (Fall 2021).
- "CS 580 Algorithmic Game Theory" (Fall 2019, Fall 2022).
- "CS 473 Algorithms" (Spring 2018).
- "CS 374 Introduction to Algorithms" (Fall 2018, Spring 2019, Spring 2020, Spring 2021, Spring 2022).

Mar 2012 - Feb 2017, *Teaching Assistant*, National Technical University of Athens

- "Algorithms and Complexity" (Fall 2016).
- "Cryptography and Number Theory" (Fall 2016).
- "Introduction to Computer Science" (Spring 2013 - Spring 2016).
- "Programming Techniques" (Spring 2012 - Spring 2016).
- "Introduction to Programming" (Fall 2012 - Fall 2016).

Experience

May 2023 - Aug 2023, *Research Intern*, Google, Mountain View

May 2022 - Aug 2022, *Software Engineering Intern*, Google, Mountain View

Jun 2021 - Dec 2022, *Software Engineer (Contractor)*, Public Benefit Enterprise of Ancient Olympia

Jun 2015 - Sep 2015, *Software Engineer (Contractor)*, Proini News